

AtCoder Educational DP Contest — Problem Reference

All tasks: 2 s, 1024 MiB. Unless stated otherwise, counts modulo $10^9 + 7$.

A — Frog 1

N stones with heights h_i . From stone i , jump to $i + 1$ or $i + 2$; cost $|h_i - h_j|$. Minimize total cost from 1 to N . **In:** $\mathbb{N}$; $\mathbf{h_1} \dots \mathbf{h_N}$
Out: minimum total cost
Constraints: $2 \leq N \leq 10^5$, $1 \leq h_i \leq 10^4$.

B — Frog 2

N stones with heights h_i . From i , jump to any existing $j \in [i + 1, i + K]$; cost $|h_i - h_j|$. Minimize cost from 1 to N . **In:** $\mathbb{N}$ K ; $\mathbf{h_1} \dots \mathbf{h_N}$
Out: minimum total cost
Constraints: $2 \leq N \leq 10^5$, $1 \leq K \leq 100$, $1 \leq h_i \leq 10^4$.

C — Vacation

For each of N days choose exactly one activity A, B, C with happiness a_i, b_i, c_i . The same activity cannot be chosen on consecutive days. Maximize total happiness. **In:** $\mathbb{N}$; then $\mathbf{a_i} \mathbf{b_i} \mathbf{c_i}$
Out: maximum happiness
Constraints: $1 \leq N \leq 10^5$, $1 \leq a_i, b_i, c_i \leq 10^4$.

D — Knapsack 1

N items, each with weight w_i and value v_i . Choose each at most once, total weight $\leq W$, maximizing total value. **In:** $\mathbb{N}$ W ; then $\mathbf{w_i} \mathbf{v_i}$
Out: maximum total value
Constraints: $N \leq 100$, $W \leq 10^5$, $w_i \leq W$, $v_i \leq 10^9$. Use 64-bit answer.

E — Knapsack 2

Same 0/1 knapsack objective, but W is large and values are small. **In:** $\mathbb{N}$ W ; then $\mathbf{w_i} \mathbf{v_i}$
Out: maximum total value
Constraints: $N \leq 100$, $W \leq 10^9$, $w_i \leq W$, $v_i \leq 1000$.

F — LCS

Given lowercase strings s, t , output any longest common subsequence. A subsequence preserves order while deleting arbitrary characters. **In:** $\mathbf{s}; \mathbf{t}$
Out: any LCS; empty string is allowed
Constraints: $1 \leq |s|, |t| \leq 3000$.

G — Longest Path

Given a directed acyclic graph, find the maximum number of edges in any directed path. **In:** $\mathbb{N}$ M ; then edges $\mathbf{x_i} \mathbf{y_i}$
Out: longest path length in edges
Constraints: $2 \leq N \leq 10^5$, $1 \leq M \leq 10^5$; edges distinct; graph is a DAG.

H — Grid 1

$H \times W$ grid of $\cdot$ and $\#$. Start (1, 1), finish (H, W) , both empty. Move only right/down through empty cells. Count paths modulo $10^9 + 7$. **In:** $\mathbb{H}$ W ; then H grid rows
Out: number of valid paths mod $10^9 + 7$
Constraints: $2 \leq H, W \leq 1000$.

I — Coins

N independent coins; coin i is heads with probability p_i . Toss all once. Find $P(\#H > \#T)$. **In:** $\mathbb{N}$; $\mathbf{p_1} \dots \mathbf{p_N}$
Out: probability; absolute error $\leq 10^{-9}$
Constraints: N odd, $1 \leq N \leq 2999$, $0 < p_i < 1$; at most 2 decimal digits.

J — Sushi

N plates initially contain $a_i \in \{1, 2, 3\}$ sushi pieces. Repeatedly choose one plate uniformly; if nonempty, eat one piece. Find expected operations until all sushi is gone. **In:** $\mathbb{N}$; $\mathbf{a_1} \dots \mathbf{a_N}$
Out: expected operations; relative error $\leq 10^{-9}$
Constraints: $1 \leq N \leq 300$.

K — Stones

Pile has K stones; allowed removals are $A = \{a_1, \dots, a_N\}$. Players alternate, first player starts; a player with no legal move loses. Determine winner under optimal play. **In:** $\mathbb{N}$ K ; $\mathbf{a_1} \dots \mathbf{a_N}$
Out: First or Second
Constraints: $N \leq 100$, $K \leq 10^5$, $1 \leq a_1 < \dots < a_N \leq K$.

L — Deque

Sequence $a_1, \dots, a_N$. Players alternately take leftmost or rightmost value and add it to their score. First maximizes $X - Y$; second minimizes it. Find optimal final $X - Y$. **In:** $\mathbb{N}$; $\mathbf{a_1} \dots \mathbf{a_N}$
Out: optimal $X - Y$
Constraints: $N \leq 3000$, $1 \leq a_i \leq 10^9$. Use 64-bit.

M — Candies

Distribute exactly K candies among N children. Child i receives an integer in $[0, a_i]$. Count valid distributions modulo $10^9 + 7$. **In:** $\mathbb{N}$ K ; $\mathbf{a_1} \dots \mathbf{a_N}$
Out: number of distributions mod $10^9 + 7$
Constraints: $N \leq 100$, $0 \leq K \leq 10^5$, $0 \leq a_i \leq K$.

N — Slimes

N adjacent slimes have sizes a_i . Repeatedly merge two adjacent slimes of sizes x, y into $x + y$, paying $x + y$, until one remains. Minimize total cost. **In:** $\mathbb{N}$; $\mathbf{a_1} \dots \mathbf{a_N}$
Out: minimum total merge cost
Constraints: $2 \leq N \leq 400$, $1 \leq a_i \leq 10^9$. Use 64-bit.

O — Matching

N men and N women; $a_{ij} \in \{0, 1\}$ indicates compatibility. Count perfect matchings using only compatible pairs, modulo $10^9 + 7$. **In:** $\mathbb{N}$; then $N \times N$ binary matrix a
Out: number of perfect matchings mod $10^9 + 7$
Constraints: $1 \leq N \leq 21$.

P — Independent Set

Given a tree, color every vertex black/white such that no edge has two black endpoints. Count valid colorings modulo $10^9 + 7$. **In:** $\mathbb{N}$; then $N - 1$ edges
Out: number of valid colorings mod $10^9 + 7$
Constraints: $1 \leq N \leq 10^5$.

Q — Flowers

Flowers are ordered left-to-right. Flower i has distinct height h_i and beauty a_i . Remove any flowers so remaining heights are strictly increasing; maximize total beauty. **In:** $\mathbb{N}$; $\mathbf{h_1} \dots \mathbf{h_N}$; $\mathbf{a_1} \dots \mathbf{a_N}$
Out: maximum total beauty
Constraints: $N \leq 2 \cdot 10^5$, $1 \leq h_i \leq N$ distinct, $1 \leq a_i \leq 10^9$. Use 64-bit.

R — Walk

Directed graph is given by adjacency matrix a . Count all directed walks of exactly K edges over every start/end pair; revisits are allowed. Answer modulo $10^9 + 7$. **In:** $\mathbb{N}$ K ; then $N \times N$ matrix a
Out: total number of length- K walks mod $10^9 + 7$
Constraints: $N \leq 50$, $1 \leq K \leq 10^{18}$, $a_{ij} \in \{0, 1\}$, $a_{ii} = 0$.

S — Digit Sum

Given huge positive integer K and integer D , count integers x with $1 \leq x \leq K$ whose decimal digit sum is divisible by D . Answer modulo $10^9 + 7$. **In:** $\mathbb{K}$; D
Out: count mod $10^9 + 7$
Constraints: $1 \leq K < 10^{10000}$, $1 \leq D \leq 100$.

T — Permutation

Given N and string s of length $N - 1$ over $<, >$, count permutations $(p_1, \dots, p_N)$ of $(1, \dots, N)$ satisfying each adjacent comparison encoded by s . Answer modulo $10^9 + 7$. **In:** $\mathbb{N}$; $\mathbf{s}$
Out: number of valid permutations mod $10^9 + 7$
Constraints: $2 \leq N \leq 3000$, $|s| = N - 1$.

U — Grouping

N rabbits; symmetric score matrix a , with $a_{ii} = 0$. Partition all rabbits into groups. For every pair $i < j$ in the same group, add a_{ij} . Maximize total score. **In:** $\mathbb{N}$; then $N \times N$ matrix a
Out: maximum total score
Constraints: $1 \leq N \leq 16$, $|a_{ij}| \leq 10^9$, $a_{ij} = a_{ji}$. Use 64-bit.

V — Subtree

Tree with modulus M . Color vertices black/white so the set of black vertices is connected. For every vertex v , count valid colorings in which v is black; output each count modulo M . **In:** $\mathbb{N}$ M ; then $N - 1$ edges
Out: N lines: answer for each $v = 1, \dots, N$ mod M
Constraints: $N \leq 10^5$, $2 \leq M \leq 10^9$.

W — Intervals

Choose any binary string $s_1 \dots s_N$. Each weighted interval (l_i, r_i, a_i) contributes a_i iff at least one position in $[l_i, r_i]$ contains 1, else 0. Weights may be negative. Maximize total score. **In:** $\mathbb{N}$ M ; then $\mathbf{l_i} \mathbf{r_i} \mathbf{a_i}$
Out: maximum total score
Constraints: $N, M \leq 2 \cdot 10^5$, $1 \leq l_i \leq r_i \leq N$, $|a_i| \leq 10^9$. Use 64-bit.

X — Tower

Block i has weight w_i , strength s_i , value v_i . Choose any subset and order it vertically. For each used block, total weight above it must be $\leq s_i$. Maximize total value. **In:** $\mathbb{N}$; then $\mathbf{w_i} \mathbf{s_i} \mathbf{v_i}$
Out: maximum total value
Constraints: $N \leq 1000$, $w_i, s_i \leq 10^4$, $v_i \leq 10^9$. Use 64-bit.

Y — Grid 2

$H \times W$ grid with exactly N listed blocked cells; (1, 1) and (H, W) are empty. Move only right/down. Count paths from start to finish modulo $10^9 + 7$. **In:** $\mathbb{H}$ W N ; then blocked cells $\mathbf{r_i} \mathbf{c_i}$
Out: number of valid paths mod $10^9 + 7$
Constraints: $H, W \leq 10^5$, $N \leq 3000$; blocked cells distinct.

Z — Frog 3

Heights satisfy $h_1 < h_2 < \dots < h_N$. From stone i , jump to any later $j > i$ with cost $(h_j - h_i)^2 + C$. Minimize total cost from 1 to N . **In:** $\mathbb{N}$ C ; $\mathbf{h_1} \dots \mathbf{h_N}$
Out: minimum total cost
Constraints: $N \leq 2 \cdot 10^5$, $C \leq 10^{12}$, $1 \leq h_1 < \dots < h_N \leq 10^6$. Use 64-bit.